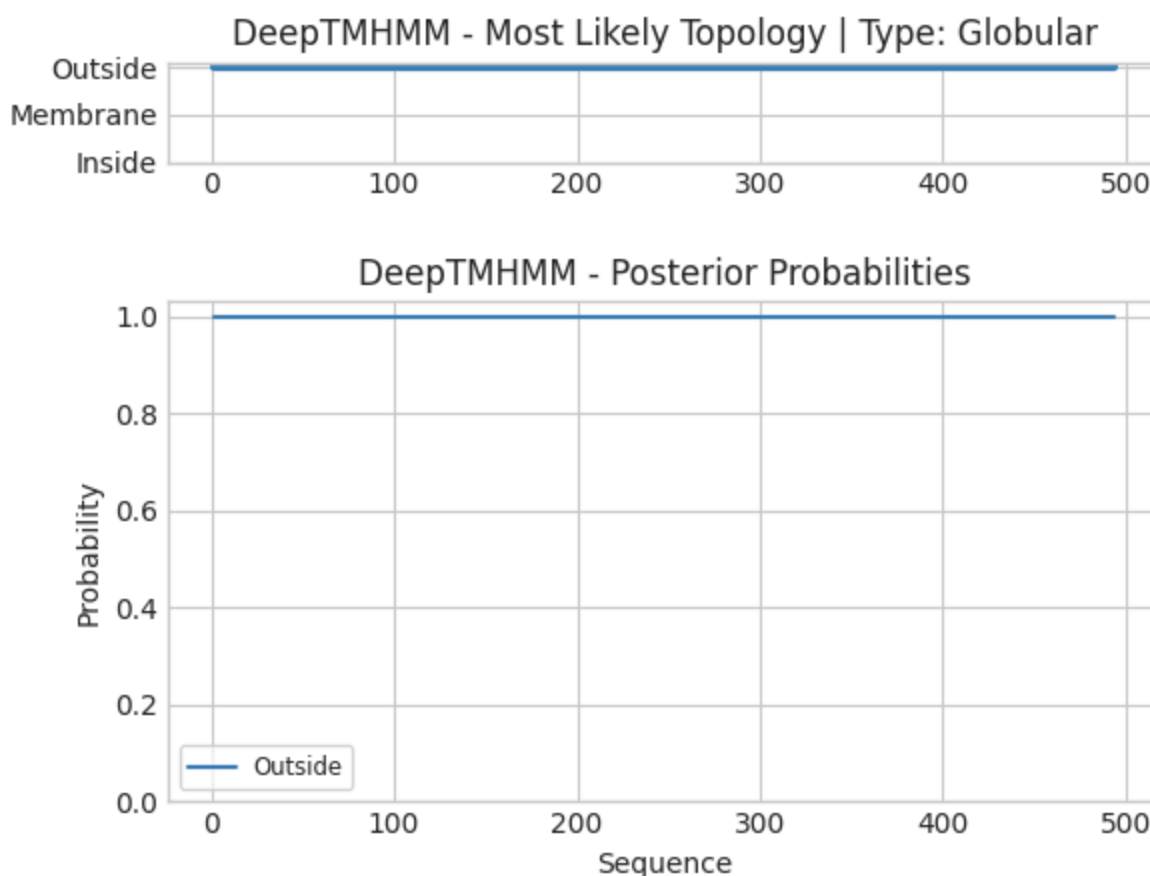


DeepTMHMM - Predictions

Predicted topologies can be downloaded in [.gff3 format](#) and [.3line format](#)



DeepTMHMM

A Deep Learning Model for Transmembrane Topology Prediction and Classification

Protein structure prediction using deep learning methods have seen several advancements within the last years. In this project, we investigate deep learning for prediction of the membrane topology of transmembrane proteins. Transmembrane proteins are relevant for drug development since they make up more than 50% of all human drug targets [3].

DeepTMHMM is currently the most complete and best-performing method for the prediction of the topology of both alpha-helical and beta-barrel transmembrane proteins. The model encodes the primary amino acid sequence by a pre-trained language model and decodes the topology by a state space model

to produce topology and type predictions at unprecedented accuracy. DeepTMHMM makes it possible to scan full proteomes in order to detect both classes of transmembrane proteins, and we anticipate our method to be very valuable for the research community.

Note: If you want to run DeepTMHMM on large datasets, then you can run it locally by following the instructions below.

Running DeepTHMMM From the Terminal

You can run DeepTMHMM directly from the terminal through the `pybiolib` package:

```
pip3 install pybiolib
biolib run DTU/DeepTMHMM --fasta input.fasta
```

Running DeepTHMMM From Python

Running DeepTMHMM from Python is showcased in the notebook below:

[Demo Colab](#)

Running DeepTHMMM Locally

By default, DeepTMHMM will be run in the BioLib Cloud.

Make sure to install docker if you want to run DeepTMHMM locally.

Docker installation instructions can be found here: <https://docs.docker.com/get-docker/>

Using GPUs locally

`nvidia-docker` is required to use your local GPUs, installation instructions can be found here:

<https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/install-guide.html>

To activate your GPUs in biolib, set the environment variable `BIOLIB_DOCKER_RUNTIME` to `nvidia`.

Running locally from Terminal

To run locally from the terminal, pass `--local` to `biolib run` .i.e.

```
biolib run --local 'DTU/DeepTMHMM:1.0.24' --fasta input.fasta
```

Running locally in Python

To run locally from python, pass `machine=local` to `cli()` .i.e.

```
biolib.utils.STREAM_STDOUT = True # Stream progress from app in real time
deepthmmm_job = deepthmmm.cli(args='--fasta query.fasta', machine='local') # Blocks until done
deepthmmm_job.save_files('result') # Saves all results to `result` dir
```

There are no sequence limitations when running locally and this allows you to process as many sequences as your machine can handle.

You can set the environment variable `BIOLIB_LOG` to `debug` for more information during execution.

Support

If you need further support, or have any questions, then don't hesitate to reach out on our [Slack](#)!

Data:

[Training Dataset](#)

[Predictions of cross validated models on dataset](#)

Reference:

Refer to the sidebar on the left for guidance on how to cite DeepTMHMM

Title: DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks

bioRxiv link: <https://www.biorxiv.org/content/10.1101/2022.04.08.487609v1>